Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
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EXERCISE 7

MD5

Code

```
import java.util.Scanner;

public class md5 {

    /* ---------- Shift amounts ---------- */
    private static final int[] SHIFT = {
        7,12,17,22, 7,12,17,22, 7,12,17,22, 7,12,17,22,
        5,9,14,20, 5,9,14,20, 5,9,14,20, 5,9,14,20,
        4,11,16,23, 4,11,16,23, 4,11,16,23, 4,11,16,23,
        6,10,15,21, 6,10,15,21, 6,10,15,21, 6,10,15,21
    };

    /* ---------- Constants ---------- */
    private static final int[] T = new int[64];

    static {
        for (int i = 0; i < 64; i++) {
            T[i] = (int)((1L << 32) * Math.abs(Math.sin(i + 1)));
        }
    }

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Input: ");
        String input = sc.nextLine();

        System.out.println("Number of characters in input: " +
input.length());

        byte[] msg = input.getBytes();
        int originalLength = msg.length;
        long bitLength = (long) originalLength * 8;
```

```java
        /* ---------- Padding ---------- */
        int padLen = (56 - (originalLength + 1) % 64 + 64) % 64;
        byte[] padded = new byte[originalLength + 1 + padLen + 8];

        System.arraycopy(msg, 0, padded, 0, originalLength);
        padded[originalLength] = (byte) 0x80;

        for (int i = 0; i < 8; i++) {
            padded[padded.length - 8 + i] = (byte)(bitLength >>> (8 * i));
        }

        /* ---------- Initial Values ---------- */
        int A = 0x67452301;
        int B = 0xefcdab89;
        int C = 0x98badcfe;
        int D = 0x10325476;

        int totalBlocks = padded.length / 64;

        for (int block = 0; block < totalBlocks; block++) {

            System.out.println("\nBlock " + (block + 1));

            int[] X = new int[16];

            for (int j = 0; j < 16; j++) {
                int idx = block * 64 + j * 4;
                X[j] = ((padded[idx] & 0xff)) |
                       ((padded[idx + 1] & 0xff) << 8) |
                       ((padded[idx + 2] & 0xff) << 16) |
                       ((padded[idx + 3] & 0xff) << 24);
            }

            int a = A, b = B, c = C, d = D;

            for (int i = 0; i < 64; i++) {

                int F, g;

                if (i < 16) {
                    F = (b & c) | (~b & d);
                    g = i;
                }
                else if (i < 32) {
                    F = (d & b) | (~d & c);
                    g = (5 * i + 1) % 16;
                }
                else if (i < 48) {
```

```
                    F = b ^ c ^ d;
                    g = (3 * i + 5) % 16;
                }
                else {
                    F = c ^ (b | ~d);
                    g = (7 * i) % 16;
                }

                int temp = d;
                d = c;
                c = b;

                int sum = a + F + T[i] + X[g];
                b = b + Integer.rotateLeft(sum, SHIFT[i]);
                a = temp;

                /* ---- Print at end of each round ---- */
                if (i == 15) {
                    System.out.printf("Round 1:\nA=%08x B=%08x C=%08x
D=%08x\n", a,b,c,d);
                }
                if (i == 31) {
                    System.out.printf("Round 2:\nA=%08x B=%08x C=%08x
D=%08x\n", a,b,c,d);
                }
                if (i == 47) {
                    System.out.printf("Round 3:\nA=%08x B=%08x C=%08x
D=%08x\n", a,b,c,d);
                }
                if (i == 63) {
                    System.out.printf("Round 4:\nA=%08x B=%08x C=%08x
D=%08x\n", a,b,c,d);
                }
            }

            A += a;
            B += b;
            C += c;
            D += d;
        }

        String finalHash = String.format("%08x%08x%08x%08x", A, B, C, D);
        System.out.println("\nFinal Hash Value: " + finalHash);
    }
}
```

Output [screenshot]

Input Test Case 1: <448 bits(<56 characters)

```
PS E:\aa vit sem6\z_Crypto-lab> javac md5.java
PS E:\aa vit sem6\z_Crypto-lab> java md5
Enter Input: arkitabarua
Number of characters in input: 11

Block 1
Round 1:
A=68fdac64 B=1e9529d1 C=81c9ee68 D=38598acc
Round 2:
A=0bd61883 B=4e4bd9f2 C=934559f6 D=1d3f723c
Round 3:
A=d6b89459 B=6a87a6fb C=7aada62c D=7c3b7a22
Round 4:
A=b1bb7868 B=35ed8f76 C=27327773 D=31f06dc1

Final Hash Value: 19009b6925bb3affbfed54714222c237
PS E:\aa vit sem6\z_Crypto-lab> java md5
```

Input Test Case 2: =448 bits(=56 characters)

```
PS E:\aa vit sem6\z_Crypto-lab> java md5
Enter Input: arkitabaruacryptographyclassarkitabaruaarkitabaruaarkita
Number of characters in input: 56

Block 1
Round 1:
A=f9d115c1 B=46616dec C=b4be789d D=b677eef2
Round 2:
A=73ae465a B=71dfb4b3 C=11edac13 D=9f2cdb04
Round 3:
A=83bb6318 B=c6cbfc93 C=67c02be8 D=d2e60f69
Round 4:
A=bc32b432 B=d76da847 C=a9e927ff D=a91ef5cc

Block 2
Round 1:
A=e011669d B=fe4e861f C=490c20ec D=b41a07de
Round 2:
A=028f9991 B=424f7ea8 C=837b3a84 D=19944877
Round 3:
A=cd6250a4 B=3573d8a3 C=e5f98c87 D=161a489a
Round 4:
A=4d39014d B=95af2b4b C=7faac5cc D=4392bc4d

Final Hash Value: 70b0d8805cea7f1bc24ecac9fce4068f
```

Input Test Case 3: >448 bits(>56 characters)

```
PS E:\aa vit sem6\z_Crypto-lab> java md5
Enter Input: arkitabaruacryptographyclassarkitabaruaarkitabaruaarkitabaruacrypto
Number of characters in input: 67

Block 1
Round 1:
A=f9d115c1 B=1d352c6c C=76836382 D=b677eef2
Round 2:
A=d946ed69 B=63456f06 C=0ad39642 D=f5cfb05a
Round 3:
A=2feaf5aa B=3ff03d5f C=d4001673 D=da52e316
Round 4:
A=956657c0 B=76d9d6c8 C=3a583f54 D=5c177410

Block 2
Round 1:
A=09b0337a B=8cfd8eba C=43253126 D=07c500ef
Round 2:
A=6d93e729 B=dd785af5 C=c4478f1c D=fb26e2ce
Round 3:
A=ece1225c B=55c83ef3 C=eeeb74e4 D=76537312
Round 4:
A=9956317b B=d1b67fe3 C=715160ce D=2b41cd0c

Final Hash Value: 9601ac3c385e023444647d20978b9592
PS E:\aa vit sem6\z_Crypto-lab>
```